HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **automation systems** for manufacturing equipment, focusing on robotics and control logic integration to enhance production throughput by approximately **15%**.
- Developed and executed comprehensive **validation strategies** and test plans for new automation deployments, ensuring adherence to stringent quality and regulatory standards.
- Led **root-cause investigations** for equipment failures and reliability issues, implementing corrective actions that reduced unscheduled downtime incidents by **22%**.
- Mentored junior engineers on standardized **automation best practices** and troubleshooting techniques, fostering a culture of continuous improvement within the team.
- Coordinated cross-functional teams, including operations and quality assurance, to ensure seamless integration and operational readiness for deployed automation solutions.
- Managed precise **documentation and configuration control** for all automation hardware and software components, maintaining compliance with industry regulations.
- Implemented **data-driven methodologies** to analyze performance metrics and optimize automation processes, contributing to a **10%** increase in overall equipment effectiveness.

SKILLS

Automation & Controls – Robotics, Control Systems, Automation Principles, Process Automation, Manufacturing Automation

Programming & Scripting – Python, SQL, PLC Programming, HMI Development

Networking – TCP/IP, Routing, Switching, Network IDS/IPS, Active Directory, Firewalls

Data Analysis & Visualization – Statistical Tools, Data Visualization, Excel, PowerPoint

Engineering Practices – Change Control, Validation, Root Cause Analysis, Continuous Improvement, Technical Writing

Cloud & Infrastructure – AWS, Docker, Linux, Jenkins, Git

PROJECTS

Automated Industrial Data Acquisition System

- Developed a Python-based system to collect real-time data from industrial sensors and equipment using **TCP/IP** protocols, integrating with a central **PostgreSQL** database.
- Implemented automated data validation checks and anomaly detection algorithms using **NumPy** and **scikit-learn** to flag deviations from normal operating parameters.
- Created automated reports and visualizations in **Power BI** to present key performance indicators to operations teams, enabling faster decision-making.

Robotic Arm Control & Integration

- Designed and programmed a **robotic arm** controller using **PLC** logic and a custom **HMI** interface for a manufacturing assembly line.
- Integrated the robotic system with existing production machinery via **REST APIs** and managed its configuration within a **Linux** environment.
- Documented the system's operational procedures and maintenance schedule, contributing to its successful deployment and adherence to **safety** standards.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)